INTERNATIONAL MONETARY FUND

FEDERATED STATES OF MICRONESIA

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staff of the International Monetary Fund and the International Development Association (IDA)[1]

Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable
Application of judgement	Yes: the forecast horizon informing mechanical risk signals is extended to 20 years to take account of an expected fiscal cliff in FY2023.

The Federated States of Micronesia (FSM) remains at high risk of debt distress under the Debt Sustainability Framework (DSF). Unless the Compact Agreement with the United States or parts of it are renewed, the FSM will face a fiscal cliff when the U.S. Compact grants amounting to 20 percent of GDP are expected to expire in FY2023. Under the baseline scenario without fiscal adjustments, the fiscal cliff would put debt on an upward trajectory starting in FY2024, with the external debt-to-GDP ratio reaching 30 percent in FY2029 and 57 percent in FY2039, and the public debt-to-GDP ratio reaching 43 percent in FY2029 and 67 percent in FY2039. As a result, the DSF thresholds on the present value of external debt-to-GDP and public debt-to-GDP ratios are projected to be breached within a 20-year horizon. While mechanical application of the DSF based on a 10-year forecast horizon would imply a moderate risk rating, the envisaged breach of the thresholds within a 20-year forecast horizon would warrant an assessment of high risk of external and overall debt distress. Lowering the risk of debt distress would require a fiscal adjustment and steadfast structural reforms to promote private sector growth. The FSM's vulnerability to climate change and weather-related natural disasters constitutes a major risk and calls for strategies to strengthen climate change resilience.

[1] This DSA has been prepared jointly by the IMF and World Bank, following the 2018 Guidance Note on the Bank-Fund Debt Sustainability Framework for Low Income Countries.

PUBLIC SECTOR DEBT COVERAGE

1. This Debt Sustainability Analysis (DSA) covers public sector debt owed by the national and state governments of the FSM. As of end-FY2018 (ending September 2018), all public sector debt is external, owed to nonresidents. Due to the lack of data, public debt covers neither loan guarantees nor non-guaranteed debt by state-owned enterprises (SOEs). While data on public-private partnerships (PPPs) are also unavailable, the institutional framework and capacity for PPPs remains underdeveloped in the FSM, suggesting that PPP capital stock is unlikely to be substantial. Against this background, the contingent liability stress test with default settings under the DSF is used to analyze public debt not covered by this DSA. The FSM uses the U.S. dollar as the legal tender and does not have a central bank.

Coverage of Public Sector Debt		
	Subsectors of the public sector	**Sub-sectors covered**
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

The country's coverage of public debt	The central, state, and local governments			
		Default	**Used for the analysis**	**Reasons for deviations from the default settings**
Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
PPP	35	percent of PPP stock	0.0	
Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

2. The FSM's public sector debt has decreased to 20 percent of GDP in FY2018 from a peak of 31 percent of GDP in FY2009. Most of the debt is concessional and is contracted with official lenders. About 70 percent of the debt is from the Asian Development Bank (AsDB), 27 percent from the U.S. Department of Agriculture (rural utilities services), and the rest (3 percent) from the European Investment Bank (EIB) and a commercial private lender. All loans except from the EIB are denominated in U.S. dollars. Total debt service reached 5.8 percent of exports in FY2018 and is expected to remain broadly stable over the medium term. Because the FSM has not borrowed debt since FY2009 and principle repayments have been proceeding, public sector debt shows a downward trajectory since FY2009.

FSM: Stock of Public Sector Debt, end-FY2018			
	In millions of U.S. dollars	As a share of total debt	In percent of GDP
Total public debt	76	100	20
External	76	100	20
Multilateral	53	70	14
ADB	52	69	14
Bilateral	22	28	6
U.S. Department of Agriculture	21	27	6
European Investment Bank (EIB)	1	1	0
Commercial	2	2	0
Telecom vendor	2	2	0
Domestic	0	0	0

Source: FSM authorities and IMF staff estimates and calculations.

BACKGROUND ON MACRO FORECASTS

3. Under the baseline projection, growth is projected to moderate to potential over the medium term, while fiscal and current account balances deteriorate in FY2024 (Box 1). Potential growth is estimated at 0.6 percent, in line with the FSM's performance over the last two decades. Inflation is projected to converge to 2 percent over the medium term, in line with the U.S. inflation. Due to the expected expiration of grants under the Compact Agreement with the United States, the fiscal and external current account balances are projected to worsen in FY2024 to deficits of 4½ percent of GDP and 4 percent of GDP, respectively. These projections are broadly in line with the previous DSA that accompanied the 2017 Article IV staff report for the FSM, with a notable exception that the post-FY2023 fiscal positions worsened. The financing mix envisages a combination of external debt, a drawdown from the FSM Trust Fund, and domestic debt. The realism tools suggest that macroeconomic and fiscal assumptions are reasonable (Figures 3 and 4).[2]

[2] Large residuals in Figure 3 reflect fiscal surpluses recorded in the past 5 years and projected through FY2023.

Box 1. Federated States of Micronesia: Key Macroeconomic Assumptions

- **GDP growth** is projected to moderate from 2.4 percent in FY2017 to 1.2–1.4 percent in FY2018–19, 0.7–0.8 percent in FY2020–21, and 0.6 percent in FY2022 and thereafter. Potential growth is estimated to be 0.6 percent, underpinned by labor productivity growth of 0.7 percent and a slight contraction in employment, in line with the FSM's performance over the last two decades. The estimate also takes account of the likely impact of natural disasters, as the FSM experienced five natural disasters during this period. Cross-country regression analyses by IMF staff[1] suggest that the FSM's high vulnerability to natural disasters warrants a moderate downward adjustment in the long-run projection for GDP growth by 0.1 percentage point.
- **Inflation** is projected to gradually pick up from 1.5 percent in FY2018 to 2 percent from FY2020 onwards, in line with inflation in the United States (the U.S. dollar is the FSM's legal tender). The GDP deflator is assumed to move in tandem with inflation.
- **The fiscal balance** is projected to be in a surplus of around 4-7 percent of GDP during FY2020–23. Despite a low tax-to-GDP ratio at about 13 percent of GDP, foreign grants, fishing license fees,[2] and other nontax revenues are expected to remain significant and fully cover government spending. Nonetheless, unless the Compact Agreement or parts of it are renewed, Compact grants from the United States amounting to 20 percent of GDP will expire in FY2023 and be replaced by investment returns accruing to the Compact Trust Fund, projected at around 11 percent of GDP in FY2024. As a result, the overall balance is projected to turn from a surplus of around 4½ percent of GDP in FY2023 to a deficit of 4½ percent of GDP in FY2024. The baseline projection assumes no fiscal consolidation efforts during FY2019–23. Due to a projected decline in distributions from the Compact Trust Fund as percent of GDP, the fiscal deficit is projected to rise further to around 6 percent of GDP in FY2029.
- **The current account balance** is projected to remain in a surplus of around 2–3 percent of GDP during FY2020–23 but turn into a deficit of 4 percent of GDP in FY2024, due to the fiscal cliff.
- **Financing mix.** With the fiscal balance remaining in surplus, no new debt disbursement is projected until FY2023. Beyond this, the paths of external and public debt are determined by projected fiscal deficits, GDP growth, interest rates for new debt, and the financing mix for fiscal gross financing needs. The financing mix envisages external project loans financing 90 percent of the fiscal deficit and a distribution from the FSM Trust Fund—a fund established by the FSM government with the aim to provide an additional revenue source to ensure long-term fiscal sustainability. The distribution is projected at $20 million per year after FY2030, as stipulated by law. The reminder is assumed to be financed by domestic borrowing from banks; such borrowing would unlikely crowd out private sector credit, given a very low loan-to-deposit ratio in the FSM.

[1] See Lee and others, 2018, "The Economic Impact of Natural Disasters in Pacific Island Countries: Adaptation and Preparedness," IMF Working Paper No.18/108.

[2] Fishing license fees during FY2019-23 are projected to remain constant in nominal terms at the FY2018 level.

FSM: Macroeconomic Assumptions

	Current DSA (2019 Article IV)				Previous DSA (2017 Article IV)			
	2017	2018	2019-23	2024-39	2017	2018	2019-23	2024-37
Real GDP (y/y growth)	2.4	1.2	0.8	0.6	2.0	1.4	0.7	0.6
CPI (y/y growth)	0.1	1.5	1.9	2.0	0.9	2.0	2.8	2.0
Fiscal balance (percent of GDP)	14.6	27.3	8.4	-5.3	8.7	8.6	8.2	-4.4
Current account balance (percent of GDP)	7.5	24.5	5.0	-4.2	3.4	3.0	2.9	-5.3

Sources: FSM authorities and IMF staff estimates and calculations.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TEST

4. The FSM's debt carrying capacity is assessed as weak. The country's Composite Indicator (CI) index of 1.69 indicates a weak debt-carrying capacity under the 2018 DSF framework, and its debt carrying capacity remains the same as the last DSA. The CI is based on a weighted average of several factors such as the country's real GDP growth, remittances, international reserves, and world growth and the Country Policy and Institutional Assessment (CPIA) score, and the calculation of the CI is based on 10-year averages of the variables, across 5 years of historical data and 5 years of projection, and the corresponding CPIA.[3] Thus, DSA thresholds applicable for the FSM are: 30 percent for the present value (PV) of external debt-to-GDP ratio, 140 percent for the PV of external debt-to-exports ratio, 10 percent for the external debt service-to-exports ratio, 14 percent for the external debt service-to-revenue ratio, and 35 percent for the PV of public debt-to GDP ratio.

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.745	1.06	63%
Real growth rate (in percent)	2.719	1.179	0.03	2%
Import coverage of reserves (in percent)	4.052	0.790	0.03	2%
Import coverage of reserves^2 (in percent)	-3.990	0.006	0.00	0%
Remittances (in percent)	2.022	4.164	0.08	5%
World economic growth (in percent)	13.520	3.559	0.48	29%
CI Score			1.69	100%
CI rating			Weak	

Applicable Thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	35

[3] The CI calculation is based on World Economic Outlook databases for April 2019 and October 2018.

5. Given the severity and frequency of natural disasters in the FSM, a tailored stress test for a natural disaster shock is added to the standard set of stress test scenarios.[4] On a natural disaster scenario, a one-off shock of 10 percentage points to debt-GDP ratio in FY2019 is assumed, with real GDP growth and exports growth lowered by 5.0 and 3.5 percentage points, respectively, in the year of the shock.[5] Shocks related to contingent liabilities are considered, using the default shock parameters, while the scenario with exchange rate depreciation is not considered as the U.S. dollar is the FSM's legal tender.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

6. Under the baseline scenario, the FSM's external debt is projected to breach a DSA threshold in the 2030s. Due to the fiscal cliff in FY2023 that would result in sizable fiscal deficits thereafter, the ratio of external debt to GDP is projected to rise from 12 percent in FY2023 to 30 percent in FY2029 and 57 percent in FY2039. As a result, the PV of external debt-to-GDP ratio breaches its threshold in FY2033, while the ratio of the PV of external debt to export and the ratios of debt service to revenue and exports remain below their thresholds throughout the projection period. The external debt path worsened compared to the previous DSA, which projected the external debt-to-GDP ratio to reach 24 percent in FY2027 and 49 percent in FY2037. This reflects the worsening of the long-run fiscal position compared to the previous DSA.

7. Stress tests point to the vulnerability of the FSM's external debt dynamics to shocks, especially related to natural disasters and a worsening of the fiscal position. The natural disaster shock would have a long-term effect on debt accumulation with the highest debt level in FY2039 among shock scenarios: the PV of external debt-to-GDP ratio rises to 41 percent in FY2029 and 83 percent in FY2039, breaching the threshold in FY2027, six years earlier than in the baseline. Among the standardized tests, a shock in the fiscal primary balance would result in the highest PV of external debt-to-GDP ratio for the 10-year horizon, breaching the threshold in FY2028.

B. Public Sector Debt Sustainability Analysis

8. Public sector debt follows closely the dynamic of external debt. Due to the fiscal cliff, fiscal deficit is expected to remain sizable from FY2024 onwards. As a result, the ratio of public debt to GDP is projected to rise from 12 percent in FY2023 to 43 percent in FY2029 and 67 percent in FY2039. Public debt rises faster than external debt, reflecting a gradually rising share of domestic debt in the government financing mix. As a result, the PV of public debt-to-GDP ratio breaches the threshold in FY2030. Like external debt, the public debt path worsened compared to the previous DSA, which projected the public debt-to-GDP ratio to reach 24 percent in FY2027 and 49 percent in FY2037. Stress tests also confirm the vulnerability of public debt dynamics to natural disasters and a worsening of the fiscal position. A negative

[4] The stress test for exchange rate depreciation is not conducted as the FSM uses the US dollar as legal tender. Because of this, the bottom-left chart in Figure 1 ("Customization of Default Settings") indicates that standardized tests are customized.

[5] The size of the shock on GDP growth (-5.0 percent), which is larger than the default value of the DSF framework (-1.5 percent), is based on cross-country regression results of Lee and others, 2018, "The Economic Impact of Natural Disasters in Pacific Island Countries: Adaptation and Preparedness," IMF Working Paper No.18/108.

growth shock can also lead to higher fiscal deficits as percent of GDP, sharply increasing the PV of public debt-to-GDP ratio to 115 percent in FY2039.

RISK RATING AND VULNERABILITIES

9. The FSM remains at high risk of external and overall debt distress. The DSA's mechanical external and overall debt-distress ratings are moderate because no DSA threshold is breached within the standard 10-year horizon. Nonetheless, following the DSA guideline, the horizon for the FSM is extended to 20 years, taking into account the fiscal cliff in FY2024 and sizable fiscal deficits thereafter, a likely structural break that would put debt on an upward trajectory barring fiscal adjustments. With one out of four external debt thresholds and the threshold for total public debt breached by end-FY2039, the FSM's external and overall debt distress risk ratings should be judged to be high. The judgement is also in line with stress test results, which indicate the vulnerability of the FSM's external and public debt paths to natural disasters and a worsening of the fiscal position. The high risk rating underscores the need to implement fiscal adjustments and develop a strategy for climate change resilience, as well as improving capacity for debt management.

10. Consideration on the FSM's access to trust and sinking funds would not alter the high risk rating. The DSA is conducted on a gross debt basis and does not net out financial assets to which the FSM government has access in the future: the Compact Trust Fund (156 percent of GDP in FY2017), the FSM Trust Fund (32 percent of GDP), and a sinking fund for public debt (about 6 percent of GDP). However, these funds should not be regarded as liquid and readily available for debt repayments: (i) under the Compact Agreement, drawdowns from the Compact Trust Fund are prohibited prior to FY2024 and limited to investment returns after FY2024; (ii) the FSM Trust Fund Act (amended in 2019 by Public Law No. 20–185) stipulates that no funds can be withdrawn from the fund before FY2030; and (iii) the size of the sinking fund, only slightly larger than the annual fiscal deficit projected post-FY2024, would not affect the thrust of the analyses presented above.

AUTHORITIES' VIEWS

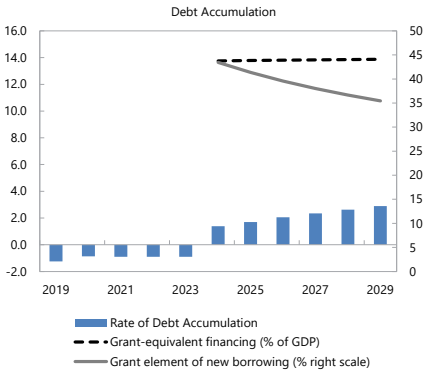
11. The authorities agreed that the FSM remains at high risk of debt distress. They supported staff's judgement to use a 20-year forecast horizon, which incorporates the FSM's country-specific prospect that U.S. Compact grants are expected to expire in FY2023 under the Compact Agreement. They also agreed that a gradual fiscal adjustment through FY2023, aimed at coping with the fiscal cliff while preserving the budget space for critical development spending, would be warranted and reduce the debt distress risk.

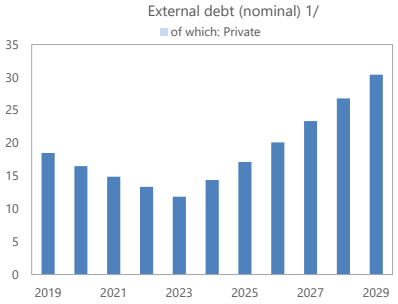
Table 1. Federated States of Micronesia: External Debt Sustainability Framework, Baseline Scenario, 2018–39

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**20.3**	**18.5**	**16.5**	**14.9**	**13.3**	**11.9**	**14.4**	**30.4**	**57.3**	**26.3**	**18.8**
of which: public and publicly guaranteed (PPG)	**20.3**	**18.5**	**16.5**	**14.9**	**13.3**	**11.9**	**14.4**	**30.4**	**57.3**	**26.3**	**18.8**
Change in external debt	-1.9	-1.8	-2.0	-1.6	-1.6	-1.5	2.5	3.6	1.0		
Identified net debt-creating flows	**-25.0**	**-15.9**	**-2.7**	**-2.5**	**-2.4**	**-2.1**	**4.2**	**4.1**	**3.9**	**4.1**	**-0.1**
Non-interest current account deficit	**-25.0**	**-16.4**	**-2.7**	**-2.8**	**-2.6**	**-2.4**	**4.0**	**3.7**	**3.0**	**4.2**	**-0.3**
Deficit in balance of goods and services	44.5	45.2	44.0	43.7	41.6	41.4	40.8	40.3	39.5	**52.2**	**41.7**
Exports	30.4	30.7	30.9	31.1	31.2	31.3	31.3	31.3	31.3		
Imports	74.9	75.9	74.9	74.8	72.8	72.7	72.1	71.7	70.8		
Net current transfers (negative = inflow)	-56.3	-48.4	-34.4	-34.1	-32.1	-31.9	-13.9	-14.3	-15.0	**-37.9**	**-24.1**
of which: official	-30.4	-30.7	-29.0	-28.4	-26.1	-25.5	-7.4	-7.4	-7.4		
Other current account flows (negative = net inflow)	-13.2	-13.2	-12.3	-12.4	-12.1	-11.9	-22.9	-22.4	-21.5	**-10.1**	**-18.0**
Net FDI (negative = inflow)	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**
Endogenous debt dynamics 2/	**0.0**	**0.4**	**-0.1**	**0.3**	**0.2**	**0.2**	**0.2**	**0.4**	**0.9**		
Contribution from nominal interest rate	0.5	0.7	0.1	0.4	0.3	0.3	0.2	0.5	1.2		
Contribution from real GDP growth	-0.3	-0.3	-0.1	-0.1	-0.1	-0.1	-0.1	-0.2	-0.3		
Contribution from price and exchange rate changes	-0.3	…	…	…	…	…	…	…	…		
Residual 3/	**23.1**	**14.1**	**0.7**	**0.9**	**0.8**	**0.6**	**-1.7**	**-0.5**	**-3.0**	**-5.0**	**1.0**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**16.4**	**14.8**	**13.6**	**12.4**	**11.2**	**10.0**	**11.1**	**20.6**	**41.3**		
PV of PPG external debt-to-exports ratio	**54.1**	**48.3**	**44.0**	**39.8**	**35.8**	**32.0**	**35.5**	**65.8**	**132.1**		
PPG debt service-to-exports ratio	**5.8**	**6.6**	**5.1**	**5.0**	**4.7**	**4.6**	**4.4**	**4.5**	**7.7**		
PPG debt service-to-revenue ratio	**3.0**	**4.1**	**4.4**	**4.4**	**4.3**	**4.2**	**3.0**	**3.2**	**5.8**		
Gross external financing need (Million of U.S. dollars)	-86.2	-54.5	-4.2	-4.8	-4.5	-3.8	23.5	25.6	35.0		
Key macroeconomic assumptions											
Real GDP growth (in percent)	1.2	1.4	0.8	0.7	0.6	0.6	0.6	0.6	0.6	**0.8**	**0.7**
GDP deflator in US dollar terms (change in percent)	1.2	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	**2.8**	**1.9**
Effective interest rate (percent) 4/	2.4	3.6	0.3	2.4	2.1	2.2	2.1	2.0	2.2	**2.6**	**2.0**
Growth of exports of G&S (US dollar terms, in percent)	3.1	3.5	3.4	3.4	2.9	2.9	2.6	2.6	2.6	**6.8**	**2.9**
Growth of imports of G&S (US dollar terms, in percent)	4.1	3.8	1.4	2.6	-0.1	2.5	1.8	2.5	2.5	**2.2**	**2.2**
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	…	…	35.5	37.2	**…**	**38.2**
Government revenues (excluding grants, in percent of GDP)	58.2	48.9	35.7	35.2	34.7	34.3	45.1	43.7	41.5	**32.3**	**41.4**
Aid flows (in Million of US dollars) 5/	142.0	147.6	143.7	144.7	136.3	137.4	62.3	69.5	87.0		
Grant-equivalent financing (in percent of GDP) 6/	…	…	…	…	…	…	13.8	13.9	13.4	**…**	**13.8**
Grant-equivalent financing (in percent of external financing) 6/	…	…	…	…	…	…	86.0	80.5	85.5	**…**	**83.2**
Nominal GDP (Million of US dollars)	372	381	391	402	413	423	434	494	640		
Nominal dollar GDP growth	2.5	2.4	2.7	2.8	2.6	2.6	2.6	2.6	2.6	**3.6**	**2.6**
Memorandum items:											
PV of external debt 7/	16.4	14.8	13.6	12.4	11.2	10.0	11.1	20.6	41.3		
In percent of exports	54.1	48.3	44.0	39.8	35.8	32.0	35.5	65.8	132.1		
Total external debt service-to-exports ratio	5.8	6.6	5.1	5.0	4.7	4.6	4.4	4.5	7.7		
PV of PPG external debt (in Million of US dollars)	61.2	56.5	53.3	49.7	46.1	42.4	48.3	101.8	264.5		
(PVt-PVt-1)/GDPt-1 (in percent)		-1.2	-0.9	-0.9	-0.9	-0.9	1.4	2.9	1.9		
Non-interest current account deficit that stabilizes debt ratio	-23.1	-14.6	-0.7	-1.2	-1.1	-0.9	1.4	0.1	2.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

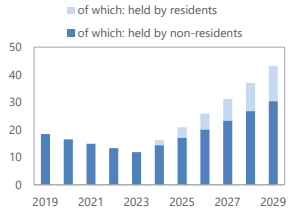
Table 2. Federated States of Micronesia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018–39

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**20.3**	**18.5**	**16.5**	**14.9**	**13.3**	**11.9**	**16.2**	**43.2**	**67.0**	**26.3**	**22.7**
of which: external debt	**20.3**	**18.5**	**16.5**	**14.9**	**13.3**	**11.9**	**14.4**	**30.4**	**57.3**	**26.3**	**18.8**
Change in public sector debt	-1.9	-1.8	-2.0	-1.6	-1.6	-1.5	4.4	6.1	1.2		
Identified debt-creating flows	**-27.6**	**-19.0**	**-7.3**	**-6.7**	**-5.7**	**-4.7**	**4.4**	**6.1**	**4.4**	**-8.2**	**-1.1**
Primary deficit	**-27.6**	**-19.2**	**-6.9**	**-6.6**	**-5.6**	**-4.7**	**4.4**	**5.8**	**4.0**	**-7.9**	**-1.1**
Revenue and grants	96.4	87.6	72.4	71.2	67.8	66.7	57.1	55.8	53.5	**70.3**	**64.0**
of which: grants	38.2	38.8	36.7	36.0	33.0	32.5	12.0	12.0	12.0		
Primary (noninterest) expenditure	68.8	68.4	65.5	64.6	62.1	62.1	61.5	61.5	57.5	**62.4**	**62.9**
Automatic debt dynamics	**0.0**	**0.2**	**-0.4**	**-0.1**	**-0.1**	**-0.1**	**-0.1**	**0.4**	**0.3**		
Contribution from interest rate/growth differential	-0.2	0.1	-0.4	-0.1	-0.1	-0.1	-0.1	0.4	0.3		
of which: contribution from average real interest rate	0.0	0.3	-0.3	0.0	0.0	0.0	0.0	0.6	0.7		
of which: contribution from real GDP growth	-0.3	-0.3	-0.1	-0.1	-0.1	-0.1	-0.1	-0.2	-0.4		
Contribution from real exchange rate depreciation	0.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**25.7**	**17.3**	**5.3**	**5.1**	**4.2**	**3.3**	**0.0**	**0.0**	**-3.1**	**7.3**	**3.2**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**16.4**	**14.8**	**13.6**	**12.4**	**11.2**	**10.0**	**12.9**	**33.4**	**51.1**		
PV of public debt-to-revenue and grants ratio	**17.1**	**16.9**	**18.8**	**17.4**	**16.5**	**15.0**	**22.6**	**59.8**	**95.5**		
Debt service-to-revenue and grants ratio 3/	**1.8**	**2.3**	**2.2**	**2.2**	**2.2**	**2.1**	**2.4**	**14.3**	**11.7**		
Gross financing need 4/	-25.8	-17.2	-5.3	-5.1	-4.1	-3.2	5.8	13.8	10.3		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	1.2	1.4	0.8	0.7	0.6	0.6	0.6	0.6	0.6	**0.8**	**0.7**
Average nominal interest rate on external debt (in percent)	2.4	3.6	0.3	2.4	2.1	2.2	2.1	2.0	2.2	**2.6**	**2.0**
Average real interest rate on domestic debt (in percent)	1.2	2.6	-1.6	0.4	0.1	0.2	0.1	6.0	6.6	**-0.1**	**2.8**
Real exchange rate depreciation (in percent, + indicates depreciation)	1.0	...	...	...	...	...	...	...	...	**-1.1**	**...**
Inflation rate (GDP deflator, in percent)	1.2	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	**2.8**	**1.9**
Growth of real primary spending (deflated by GDP deflator, in percent)	8.6	0.9	-3.6	-0.7	-3.2	0.5	-0.3	0.6	-0.3	**2.6**	**-0.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-25.7	-17.4	-4.9	-5.0	-4.1	-3.2	0.1	-0.4	2.8	**-14.8**	**-3.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

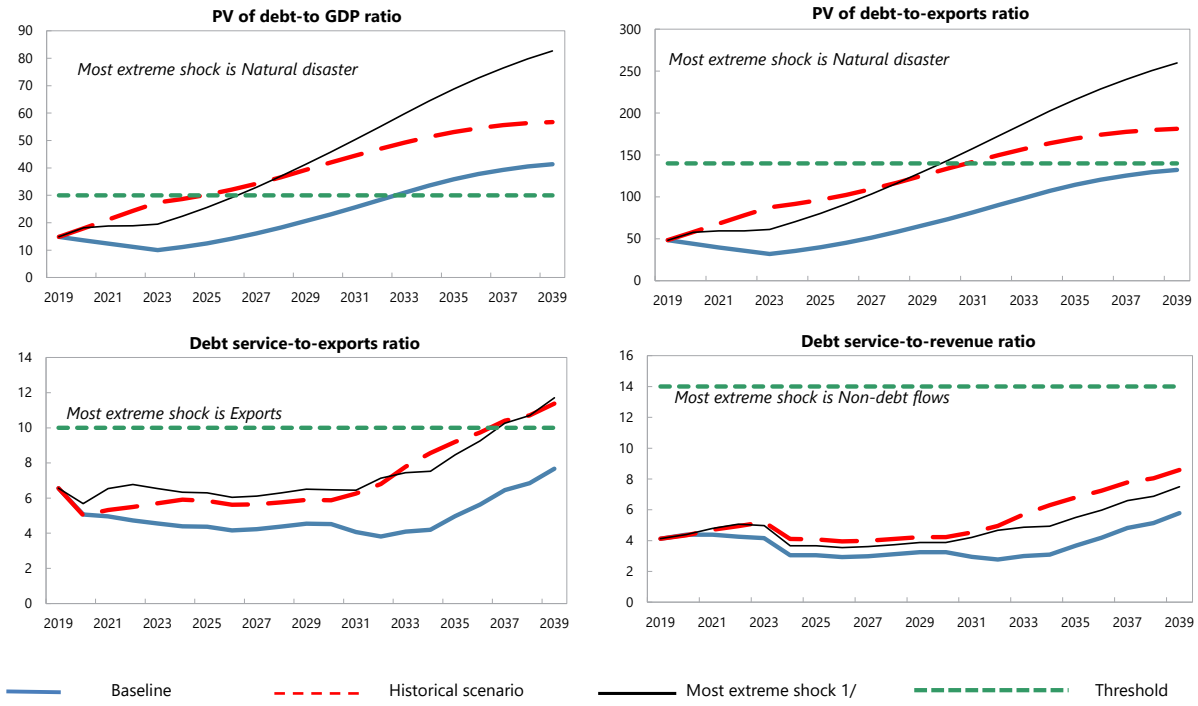
Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/



Sources: Country authorities; and staff estimates and projections.
1/ Coverage of debt: The central, state, and local governments. Definition of external debt is Residency-based.
2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.
4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.
5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.
6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Federated States of Micronesia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2019–39



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	
Tailored Tests		
Combined CLs	No	
Natural Disasters	No	Yes
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.1%	2.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	10	10

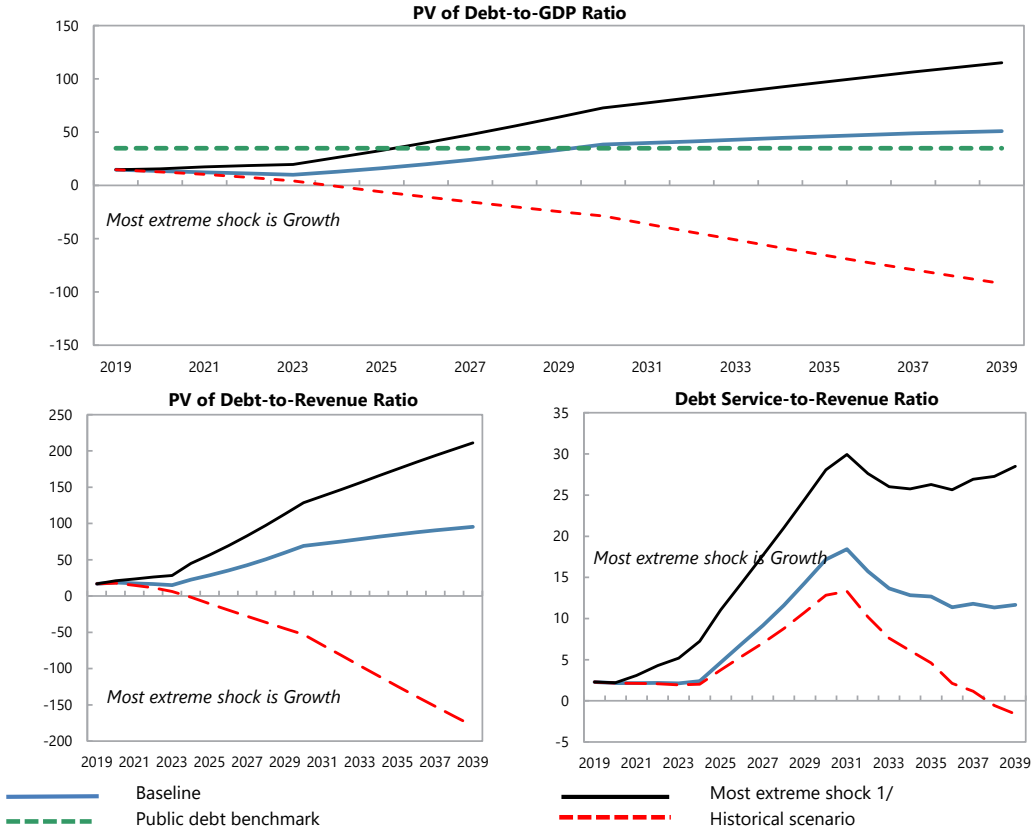
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these on off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Federated States of Micronesia: Indicators of Public Debt Under Alternative Scenarios, 2019–39



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	50%	50%
Domestic medium and long-term	17%	17%
Domestic short-term	32%	32%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.1%	2.1%
Avg. maturity (incl. grace period)	29	29
Avg. grace period	10	10
Domestic MLT debt		
Avg. real interest rate on new borrowing	7.0%	7.0%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	5.0%	5.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Federated States of Micronesia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019–39

(In percent)

	Projections 1/											
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2039
PV of debt-to GDP ratio												
Baseline	15	14	12	11	10	11	12	14	16	18	21	**41**
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	15	18	21	24	27	29	**30**	**32**	**34**	**37**	**39**	**57**
B. Bound Tests												
B1. Real GDP growth	15	14	13	12	11	12	14	15	18	20	22	**45**
B2. Primary balance	15	16	19	19	19	21	24	26	28	**31**	**33**	**54**
B3. Exports	15	16	19	18	17	18	20	22	24	26	29	**48**
B4. Other flows 3/	15	18	21	20	19	21	22	24	26	28	**31**	**48**
B5. Depreciation	15	14	12	11	10	11	12	14	16	18	21	**41**
B6. Combination of B1-B5	15	18	19	18	17	18	19	21	23	26	28	**47**
C. Tailored Tests												
C1. Combined contingent liabilities	15	16	15	15	14	15	17	19	21	23	26	**48**
C2. Natural disaster	15	18	19	19	19	22	26	29	**33**	**37**	**41**	**83**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	30	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio												
Baseline	48	44	40	36	32	35	40	45	51	58	66	132
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	48	58	68	78	87	92	97	102	109	117	125	**181**
B. Bound Tests												
B1. Real GDP growth	48	44	40	36	32	35	40	45	51	58	66	132
B2. Primary balance	48	53	61	62	61	69	75	82	90	98	107	**174**
B3. Exports	48	57	76	71	67	71	77	84	93	102	112	**186**
B4. Other flows 3/	48	59	69	65	62	66	70	76	83	90	98	**153**
B5. Depreciation	48	44	40	36	32	35	40	45	51	58	66	132
B6. Combination of B1-B5	48	60	58	61	57	61	66	72	80	88	96	**163**
C. Tailored Tests												
C1. Combined contingent liabilities	48	51	50	47	45	49	54	60	67	74	82	**152**
C2. Natural disaster	48	58	59	60	61	70	80	91	104	116	130	**260**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	140	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio												
Baseline	7	5	5	5	5	4	4	4	4	4	5	8
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	7	5	5	5	6	6	6	6	6	6	6	11
B. Bound Tests												
B1. Real GDP growth	7	5	5	5	5	4	4	4	4	4	5	8
B2. Primary balance	7	5	5	5	5	5	5	5	5	5	6	11
B3. Exports	7	6	7	7	7	6	6	6	6	6	7	12
B4. Other flows 3/	7	5	5	6	5	5	5	5	5	5	5	10
B5. Depreciation	7	5	5	5	5	4	4	4	4	4	5	8
B6. Combination of B1-B5	7	5	6	6	6	6	6	5	5	6	6	10
C. Tailored Tests												
C1. Combined contingent liabilities	7	5	5	5	5	5	5	5	5	5	5	8
C2. Natural disaster	7	5	6	5	5	5	6	5	6	6	6	11
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	10	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio												
Baseline	4	4	4	4	4	3	3	3	3	3	3	6
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	4	4	5	5	5	4	4	4	4	4	4	9
B. Bound Tests												
B1. Real GDP growth	4	5	5	5	5	3	3	3	3	3	4	6
B2. Primary balance	4	4	5	5	5	4	4	4	4	4	4	8
B3. Exports	4	4	5	5	5	4	4	3	4	4	4	7
B4. Other flows 3/	4	4	5	5	5	4	4	4	4	4	4	8
B5. Depreciation	4	4	4	4	4	3	3	3	3	3	3	6
B6. Combination of B1-B5	4	4	5	5	5	4	4	3	4	4	4	7
C. Tailored Tests												
C1. Combined contingent liabilities	4	4	5	5	4	3	3	3	3	3	4	6
C2. Natural disaster	4	4	5	5	5	4	4	4	4	4	4	8
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	14	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Federated States of Micronesia: Sensitivity Analysis for Key Indicators of Public Debt, 2019–39

(In percent)

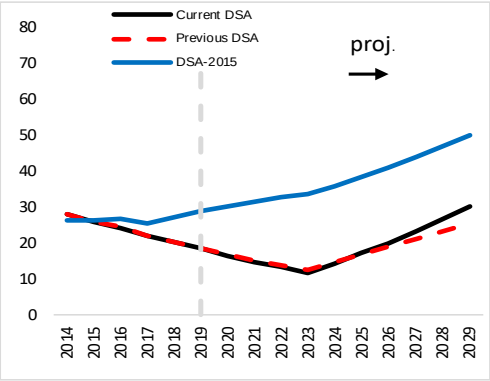
	Projections 1/											
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2039
PV of Debt-to-GDP Ratio												
Baseline	15	14	12	11	10	13	16	20	24	28	33	**51**
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	15	13	11	8	4	-1	-6	-11	-15	-20	-24	-92
B. Bound Tests												
B1. Real GDP growth	15	16	17	19	20	26	33	**40**	**48**	**56**	**64**	**115**
B2. Primary balance	15	21	26	25	24	27	30	34	**38**	**42**	**47**	**66**
B3. Exports	15	16	19	18	17	20	23	27	31	**35**	**40**	**56**
B4. Other flows 3/	15	18	21	20	19	22	26	30	34	**38**	**43**	**58**
B5. Depreciation	15	14	12	11	10	13	16	20	24	28	33	**51**
B6. Combination of B1-B5	15	18	17	14	13	16	20	24	28	33	**38**	**56**
C. Tailored Tests												
C1. Combined contingent liabilities	15	19	18	17	15	18	22	25	29	34	**39**	**58**
C2. Natural disaster	15	24	24	24	24	29	34	**40**	**46**	**53**	**60**	**98**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	35	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio												
Baseline	17	19	17	16	15	23	29	35	43	51	60	95
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	17	18	15	11	6	(2)	(11)	(19)	(28)	(36)	(45)	(179)
B. Bound Tests												
B1. Real GDP growth	17	21	23	26	28	45	57	70	83	98	113	211
B2. Primary balance	17	29	37	37	36	46	52	59	67	76	85	123
B3. Exports	17	22	26	26	25	34	40	47	55	63	72	104
B4. Other flows 3/	17	25	30	30	29	39	45	52	60	69	78	108
B5. Depreciation	17	19	17	16	15	23	29	35	43	51	60	95
B6. Combination of B1-B5	17	25	24	20	19	28	35	42	49	58	67	104
C. Tailored Tests												
C1. Combined contingent liabilities	17	27	25	25	23	32	38	45	52	61	70	107
C2. Natural disaster	17	32	33	35	35	50	60	70	81	93	106	182
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio												
Baseline	2	2	2	2	2	2	5	7	9	12	14	12
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2019-2029 2/	2	2	2	2	2	2	4	5	7	9	11	(2)
B. Bound Tests												
B1. Real GDP growth	2	2	3	4	5	7	11	14	18	21	25	28
B2. Primary balance	2	2	7	9	8	9	9	10	12	14	17	16
B3. Exports	2	2	2	2	2	3	5	7	10	12	15	13
B4. Other flows 3/	2	2	2	3	3	3	5	7	10	12	15	13
B5. Depreciation	2	2	2	2	2	2	5	7	9	12	14	12
B6. Combination of B1-B5	2	2	2	4	3	4	7	9	12	14	17	15
C. Tailored Tests												
C1. Combined contingent liabilities	2	2	6	4	5	5	6	8	10	13	15	12
C2. Natural disaster	2	2	8	6	8	8	10	13	16	18	21	21
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

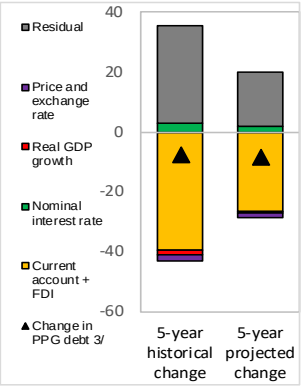
Figure 3. Federated States of Micronesia: Drivers of Debt Dynamics – Baseline Scenario

External debt

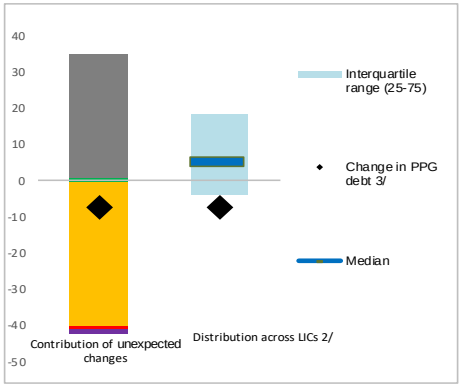
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

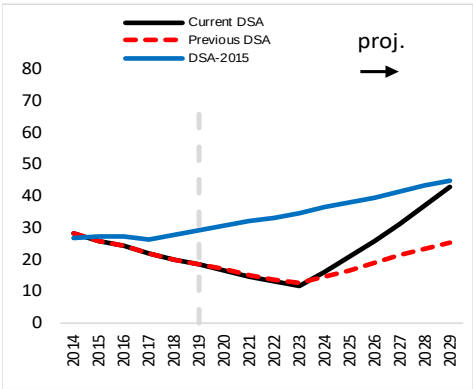


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

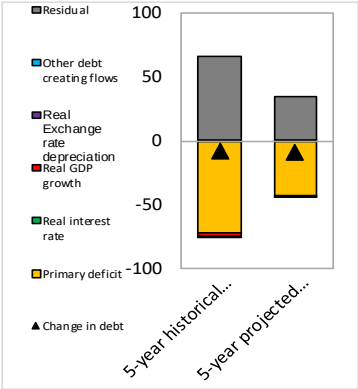


Public debt

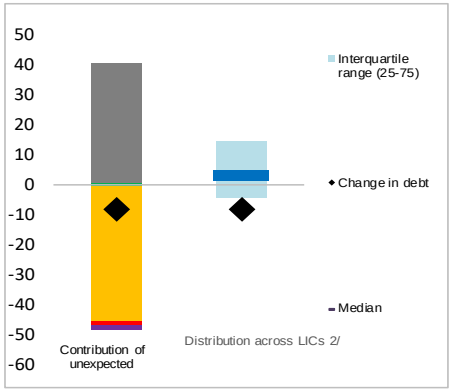
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



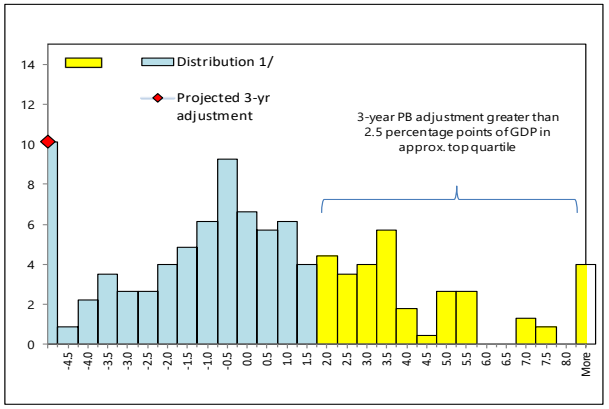
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

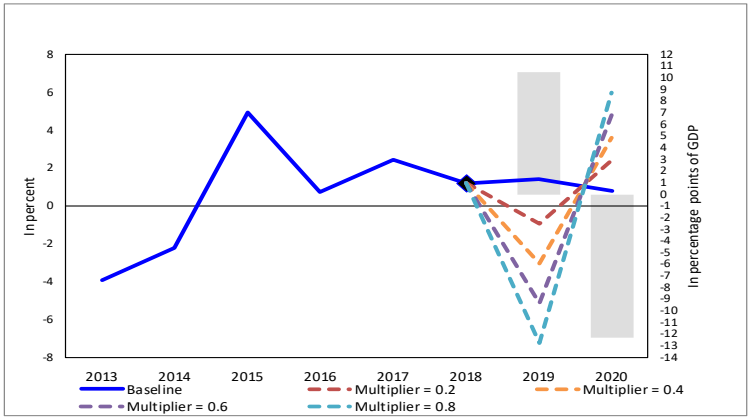
Figure 4. Federated States of Micronesia: Realism Tools 1/

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



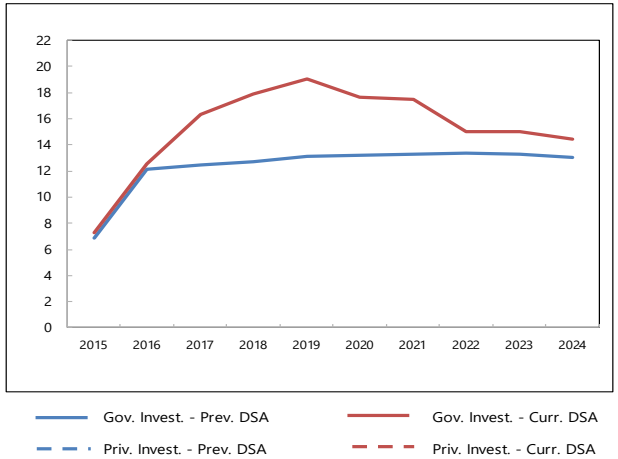
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



1/ The data needed to conduct the investment growth realism tool is not available for the DSA exercise.